

ChE 210 – Material and Energy Balance –Fall 2025

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problem #1)

SLO 1 Performance Indicator (PI)	(3) Satisfactory	(2) Developing	(1) Unsatisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems
4. Translates academic theory into engineering applications	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality	Some gaps in understanding the application of theory to the problem and expects theory to predict reality	Does not appear to grasp the connection between theory and the problem

Averages:

Performance Indicators: (1) 2.16 (2) 1.94 (4) 1.84 Overall Average: 1.98

PI1 is the strongest score by a clear margin. This score was evaluated on students' ability to carry out numerical calculations for material balance equations for a multicomponent, multiunit operations process with reaction. The calculations were made during the final exam using pocket calculators. The students who struggle with this task used calculations that were not based on concepts of mass conservation or made numerical errors using their calculators. Some students do fail to understand that ad hoc combinations of values appearing in the problem statement will not satisfy conservation of mass. Although this is a central principle of the course, it can be re-enforced by more specific exercises in the discussion portion of the course. Also, a problem can be added to the homework giving calculation exercises derived from the type seen in material balances.

The PI2 is the second highest average. Here we focus on the solution of the material balance equations after they are set up. This usually requires algebra for steady state problems and differential equations for transient or unsteady equations. Once the student has developed the material balance equations correctly, can the student correctly solve them? Students arrive in ChE 210 with inadequate math skills, even at the level of solving systems of algebraic equations. Solving linear and non-linear algebraic equations is a prerequisite skill for the course; however, students' proficiency has decreased recently. Most students will have taken CHE 205 which is a computer applications course that covers solving systems of linear and non-linear algebraic equations. An additional discussion exercise will be developed to demonstrate setting up balances for these multi-unit operations processes.

Finally, PI4 is the lowest and assesses being able to set up the material balance equations for the chosen systems. For this example, students needed to define systems around unit operations and groups of unit operation. For a system, they must identify the input and output stream variable for the four species being processed. There is some mixed success for students doing this. It is perhaps the most difficult new skill they need to acquire, that is, engineering analysis. Students need to recognize whether a system is well posed or under specified. This problem is completely linear and can be solved with linear equations only. The students who had trouble with the problem struggled with setting up a set of solvable equations. In the future, more examples of this type will be added to the discussion section for practice.

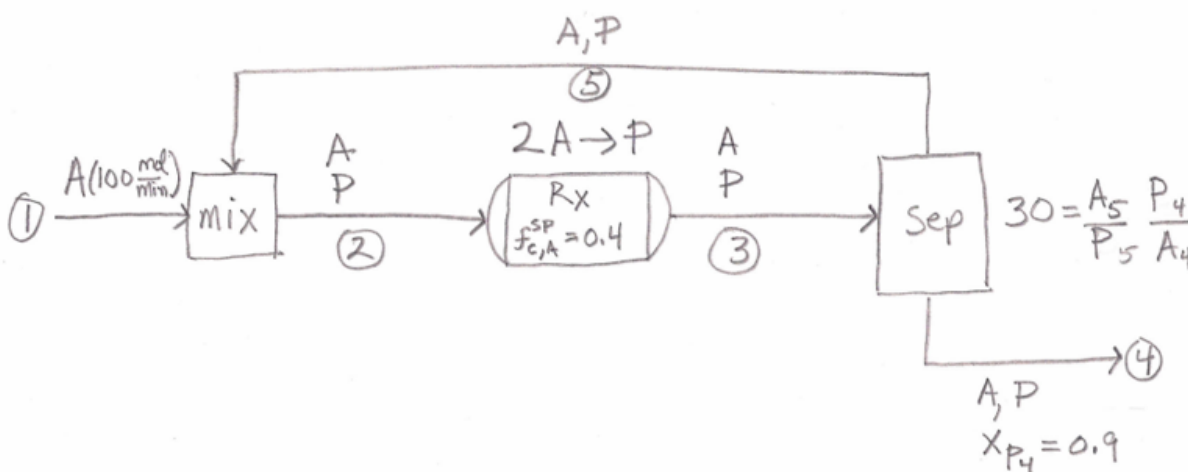
- (1) (50 pts.) Consider the block flow diagram below. Notice that the streams are labeled 1 through 5. Species molar flow rates are given by the species symbol with the stream number as a subscript. The fresh feed contains only species A. This stream is mixed with a recycle stream containing A and P where P is the reaction product. The mixture enters a reactor in which one reaction takes place:



The reactor converts 40% of A entering the reactor. The outlet stream from the reactor goes to a separator. Both streams leaving the separator are in equilibrium containing some A and P such that:

$$\alpha_{AB} = \frac{A_5 P_4}{P_5 A_4}$$

where the separation factor $\alpha_{AB} = 30$. Stream 5 is the stream enriched in A and is recycled back to the reactor. Stream 4 is enriched in the product P such that it is 90 mol% P. The fresh feed has species flow rates of $A_1 = 100$ mol A/min.



- (a) Find all of the species molar flow rates not given in the problem statement. Fill in the table below with the appropriate species flow rates in mole/min to receive full credit. (45 pts.)

Species↓/Stream→	1	2	3	4	5
A	100				
P	----				

- (b) Calculate the overall fractional conversion based on species A (5 pts.)

$$f_{c,A}^{ov} = \underline{\hspace{2cm}}$$